

FINANCIAL MARKETS IN CONTINUOUS TIME

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1. ANOTHER LOOK AT DISCRETE TIME STOCHASTIC PROCESSES

Let $(\Omega, \mathcal{F}, P)$ be a probability space. We usually think of a discrete random process as a finite collection of measurable (adapted to $\mathcal{F}$) functions $\{S_n\}_{n \leq N} : \Omega \rightarrow \mathbb{R}$.

Alternatively we can think of this random process as a **probability measure** μ on $(\mathbb{R}^N, \mathcal{B})$ for which we determine its joint **distribution**. That is, if $A = [a_1, b_1] \times \cdots \times [a_N, b_N]$ then we think of $\mu(A) = \mathbb{P}(S_n \in [a_n, b_n] : n \leq N)$. The process is then given by the coordinate maps $S_n = \omega_n$.

Adapted processes. Let $(\mathbb{R}^N, \mathcal{B}, \mu)$ and let the coordinate map define a process $B_n = \omega_n$. Given any collection of measurable functions $\{\phi_n\} : \mathbb{R}^n \rightarrow \mathbb{R}$ we can define a **predictable process**

$$T_n = \phi_n(\omega_1, \dots, \omega_{n-1}).$$

Remark 1. This is a particular class of process from the more general defined by $T_n = \phi_n(\omega)$ (which we could also have written $\phi(\omega; n)$ in analogy with the usual notation in continuous time).

Martingale transform. Now suppose that B_n is a martingale and we have a predictable process $T_n = \phi(\omega; n)$. We can define the **martingale transform**

$$(T \cdot B)_n = \sum_{j=1}^n T(\omega; j)(B_j - B_{j-1})(\omega).$$

Remark 2. This definition for predictable processes is well suited for defining the value process of a portfolio. In that case the predictable process T is our strategy.

Remark 3. Compare to the definition of Itô's integral below, in that case the process is *adapted*, i.e. $T_n = \phi_n(\omega_1, \dots, \omega_n)$ so we instead write

$$[T \cdot B]_n = \sum_{j=0}^{n-1} T(\omega, j)(B_{j+1} - B_j)(\omega).$$

We observe the following.

Lemma 1.1. *Let T be adapted. Then, $[T \cdot B]$ is a martingale. Moreover,*

$$\mathbb{E}_\mu([T \cdot B]_n)^2 = \sum_{j=0}^{n-1} T^2(\omega; j) \mathbb{E}_\mu((B_{j+1} - B_j)^2 | \mathcal{F}_j)(\omega)$$

In, particular, if $\mathbb{E}_\mu((B_{j+1} - B_j)^2 | \mathcal{F}_j) = t_{j+1} - t_j$ we have

$$(\text{It\^o's isometry}) \quad \mathbb{E}_\mu((T \cdot B)_n)^2 = \sum_{j=0}^{n-1} T^2(\omega; j)(t_{j+1} - t_j).$$

Proof. Simply write the double sum and use that the increments are orthogonal. $\square$

2. CONTINUOUS TIME STOCHASTIC PROCESSES

We think of (continuous time) stochastic process as **probability measures** on $\mathbb{R}^{[0,T]}$, i.e. the on the **space of functions** $f : [0, T] \rightarrow \mathbb{R}$.

2.1. Construction of Brownian motion.

Definition 2.1. We say that the random process B_t is a **Brownian motion** if:

- $B_0(\omega) = 0$ a.s.
- (*Continuity*): $t \mapsto B_t(\omega)$ is continuous a.s.
- (*Stationary independent Gaussian increments*): For all $s < t$ the random variable $B_t - B_s$ is a normal random variable with zero mean and variance $t - s$ and is independent of $\mathcal{F}_s = \sigma(B_u, u \leq s)$.

We now show that this process actually exists. Before doing that let us point out that we think of Brownian motion as a **probability measure** on the probability space $(C, \mathcal{C})$ where $C = \{\text{continuous } \omega : [0, \infty) \rightarrow \mathbb{R}\}$ and $\mathcal{C}$ is the σ -algebra generated by the coordinate maps $t \mapsto \omega(t)$. What we ask in the definition corresponds to this measure assigning probability zero to non-continuous functions and fixing the finite dimensional distributions.

Remark 4. Given the above probability measure on $(C, \mathcal{C})$ the random process B_t is simply given by $B_t(\omega) = \omega(t)$.

Theorem 2.2. *The Brownian motion random process exists.*

Proof. The proof follows from the **Kolmogorov extension theorem**. In order to deal with continuity this has to be done in a slightly indirect way.

- **Step 1:** Check consistency of finite-dimensional distributions (see Durrett).
- **Step 2:** Consider the (countable) product space $\Omega_q = \{\omega : \mathbb{Q}_{dyad} \rightarrow \mathbb{R}\}$ and let $\mathcal{F}_q$ denote the product (cylinder) σ -algebra. It is then possible (see Durrett) to construct a (unique) probability measure ν on Ω_q which:
 - Assigns probability one to paths $\omega : \Omega_q \rightarrow \mathbb{R}$ which are uniformly continuous on $\mathbb{Q}_2 \cap [0, T]$ (for any $T < \infty$) and
 - Has the given finite dimensional distributions, i.e.

$$\nu\{\omega(t_i) \in A_i\} = \int_{A_1} dx_1 \cdots \int_{A_n} dx_n \prod_{i=1}^n p_{t_i - t_{i-1}}(x_{i-1}, x_i)$$

$$\text{where } p_t(a, b) = \frac{1}{\sqrt{2\pi t}} \exp(-(b - a)^2 / 2t).$$

- **Step 3:** Extend to a process defined for $t \in \mathbb{R}$. Then, let $\tilde{\Omega}_q = \{\omega \in \Omega_q : \omega \text{ is uniformly continuous}\}$. Define the measure $P = \nu \circ \psi^{-1}$ on $(C, \mathcal{C})$ using that the map

$$\psi : \tilde{\Omega}_q \rightarrow C$$

that assigns the (unique) continuous extension is measurable (see Durrett).

□

It is not difficult to prove the following.

Proposition 2.3. *The function $t \mapsto B_t(\omega)$ is not differentiable a.s.*

In the following we let $(\Omega, \mathcal{F}, P)$ be the classical Wiener space (see later in this file), let $\mathcal{F}_t$ be the sigma-algebra generated by the coordinate maps and let $\mathcal{V}$ be the set of functions

$$f : (t, \omega) \in [0, \infty) \times \Omega \rightarrow \mathbb{R}$$

such that

- (1) f is $\mathcal{B} \times \mathcal{F}$ measurable,
- (2) $\omega \mapsto f$ is $\mathcal{F}_t$ measurable (adapted),
- (3) $\mathbb{E}(\int_0^\infty f^2 dt) < \infty$.

2.2. The Ito integral. The material in this section is well explained in Oksendal's book. We construct the Ito integral in three steps:

-) **Definition for elementary functions:** We say a function ϕ is elementary if

$$\phi(t, \omega) = \sum e_j(\omega) \mathbb{1}_{[t_j, t_{j+1}]}$$

where $e_j(\omega)$ is $\mathcal{F}_{t_j}$ -measurable. For elementary functions we define

$$\int_0^T \phi(t, \omega) dB_t = \sum_j e_j(\omega) (B_{t_{j+1}} - B_{t_j})(\omega).$$

-) **Ito isometry:** If ϕ is bounded and elementary then

$$\mathbb{E} \left(\int_0^T \phi(t, \omega) dB_t \right)^2 = \mathbb{E} \int_0^T \phi(t, \omega)^2 dt$$

-) **Extension to $f \in \mathcal{V}$:** This is done via an approximation procedure which exploits the Ito isometry. First we construct a sequence $\{\phi_n\} \subset \mathcal{V}$ such that

$$\mathbb{E} \int_0^T (f - \phi_n)^2 dt \rightarrow 0.$$

We then define (note the limit exists as an element of L^2)

$$I(f)[\omega] = \int_0^T f(t, \omega) dB_t(\omega) := \lim_{n \rightarrow \infty} \int_0^T \phi_n(t, \omega) dB_t(\omega).$$

2.3. Ito's formula. We now give Ito's lemma, which can be seen as a generalization of the fundamental theorem of calculus to functions of Brownian motion.

Lemma 2.4 (Ito's lemma). *Let $f(t, x) \in C^2$. Then,*

$$f(t, B_t) - f(0, B_0) = \int_0^t \left(\partial_t f(s, B_s) + \frac{1}{2} \partial_{x^2}^2 f(s, B_s) \right) ds + \int \partial_x f(s, B_s) dB_s.$$

Remark 5. The appearance of the second term is due to the fact that B_t has quadratic variation different from zero (see below).

Proof. The complete proof can be found in Durrett. Here we just deal with the second term. Also, we only analyze f independent of t so, if $s < t$, by the mean value theorem

$$f(B_t) - f(B_s) = \partial_x f(B_s)(B_t - B_s) + \frac{1}{2} \partial_{x^2}^2 f(c(B_s, B_t))(B_t - B_s)^2$$

where $c(B_s, B_t)$ is a number between B_s and B_t . We want to compute

$$\sum_i \partial_{x^2}^2 f(c(B_i, B_{i-1}))(B_i - B_{i-1})^2 \quad (2.1)$$

To that end we show that $\sum_i (B_i - B_{i-1})^2 \rightarrow t$ a.s. That is, the cumulative distribution (separate monotone increasing and monotone decreasing see construction of Lebesgue-Stieltjes integral) of the measure converges to t . Hence the measure converges weakly to dt . To check (2.1) we show that

$$\mathbb{E}(Q(B_t) - t)^2 = 0$$

where the **quadratic variation** operator Q is defined by

$$Q(f) = \sup_{\mathcal{P}} \sum_{i \in \mathcal{P}} (f(t_i) - f(t_{i-1}))^2.$$

To that end we write

$$\begin{aligned} \mathbb{E} \sum_i ((B_{t_i} - B_{t_{i-1}})^2 - (t_i - t_{i-1}))^2 &= \sum_i \mathbb{E} ((B_{t_i} - B_{t_{i-1}})^2 - (t_i - t_{i-1}))^2 \\ &= \sum_i \mathbb{E} (B_{t_i} - B_{t_{i-1}})^4 - 2\mathbb{E} (B_{t_i} - B_{t_{i-1}})^2 (t_i - t_{i-1}) + (t_i - t_{i-1})^2 \end{aligned}$$

Now use that, since $B_{t_i} - B_{t_{i-1}}$ is normally distributed with variance $t_i - t_{i-1}$ we have

$$\mathbb{E} (B_{t_i} - B_{t_{i-1}})^2 = t_i - t_{i-1} \quad \mathbb{E} (B_{t_i} - B_{t_{i-1}})^4 = (t_i - t_{i-1})^2$$

and our claim follows easily. $\square$

2.4. The martingale representation theorem. It can be shown by means of the Doob martingale inequality that, if $f \in \mathcal{V}$, then $\int_0^T f(t, \omega) dB_t(\omega)$ is an $\mathcal{F}_t$ -martingale. The converse to this statement is key to establish completeness of the Black-Scholes model.

Theorem 2.5. *Suppose that M_t is a $\mathcal{F}_t$ -martingale and $M_t \in L^2$. Then, there exists a unique $g \in \mathcal{V}$ such that*

$$M_t(\omega) = \mathbb{E} M_t + \int_0^t g(s, \omega) dB_s(\omega)$$

Sketch of the proof. We use that:

- The linear span of the set of functions of the form $Y = \exp(\int_0^T h dB_t - \frac{1}{2} \int_0^T h^2 dt)$ with $h(t) \in L^2[0, T]$ is dense in $L^2(\mathcal{F}_T)$.
- For Y as above $Y = 1 + \int_0^T Y_t(s, \omega) h(s) dB_t$ where $Y_t = \exp(\int_0^t h dB_s - \frac{1}{2} \int_0^t h^2 ds)$.
- Approximate $F \in L^2(\mathcal{F}_T)$ by linear combination of Y as above and obtain a sequence $\{f_n\}$ such that

$$F_n = \mathbb{E}F_n + \int_0^T f_n dB_t.$$

- Use the Ito isometry to show that f_n form a Cauchy sequence in $L^2(\mathcal{F}_T)$.

□

2.5. Cameron-Martin-Girsanov's theorem. The following result is of key importance to construct an equivalent martingale measure in the case where the risky asset is driven by a Brownian motion with drift.

Theorem 2.6. *Suppose that the random process B_t is a Brownian motion under the probability measure P and let Q_θ be the probability measure defined by*

$$\frac{dQ_\theta}{dP} = Z_\theta := \exp(\theta B_t - \theta^2 t/2).$$

Then, under Q_θ the random process B_t is a Brownian motion with drift $-\theta$.

Remark 6. Notice that Z_θ is a random variable which satisfies $\mathbb{E}_P(Z_\theta) = 1$. Thus, Q_θ is a probability measure (it is defined by $Q_\theta(A) = \mathbb{E}_P(Z_\theta \mathbb{1}_A)$).

Proof. First note that if we denote by $U = B_1$ we have that

$$\begin{aligned} Q_\theta(U \leq x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp(\theta u - \theta^2/2) \exp(-u^2/2) du \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp(-(u - \theta)^2/2) du \\ &= P(U - \theta \leq x). \end{aligned}$$

(so the result seems plausible). To actually check the result we just need to check that the finite dimensional distributions of B_t under Q_θ and of $B_t - \theta t$ under P agree. We can do this using for example characteristic functions. □

2.6. The classical Wiener space. The classical Wiener space $(\Omega, \mathcal{F}, P)$ is the probability space where

- $\Omega = C = \{\text{continuous } \omega : [0, \infty) \rightarrow \mathbb{R}\}$.
- $\mathcal{F} = \mathcal{C}$ is the σ -algebra generated by the coordinate maps $t \mapsto \omega(t)$ (cylinder σ -algebra).
- $P = \nu$ is the unique measure on $(\Omega, \mathcal{F})$ for which the coordinate process is Brownian motion.

3. ONE-DIMENSIONAL MARKET ON THE CLASSICAL WIENER SPACE

We consider the classical Wiener space $(\Omega, \mathcal{F}, P)$ and consider a market consisting of

- S_t^0 is a **bond**, whose dynamics is given by the ODE

$$dS_t^0 = rS_t^0 dt$$

- S_t^1 is a **risky asset** whose dynamics is given by the SDE

$$dS_t^1 = S_t^1(\mu dt + \sigma dB_t).$$

3.1. Completeness of the market model. Completeness of the market model follows from the martingale representation theorem.

3.2. Equivalent martingale measure and strategies. We first construct the **equivalent martingale measure** P_μ using Girsanov's theorem. Namely we choose

$$\frac{dP_\mu}{dP} = \exp(-\theta B_t - \frac{1}{2}\theta^2 t) \quad \theta = \frac{\mu - r}{\sigma}.$$

Under the measure P_μ , the process $B_t^\mu = \frac{\mu - r}{\sigma}t + B_t$ is the standard Brownian measure.

Lemma 3.1. *Under the measure P_μ the discounted process $\tilde{S}_t^1 = e^{-rt}S_t^1$ satisfies the SDE*

$$d\tilde{S}_t^1 = \tilde{S}_t^1 \sigma dB_t^\mu.$$

3.2.1. The value process. Given a strategy (adapted process) $\phi = (H_t^0, H_t^1)$ we define the associated value process

$$V_t(\phi) = H_t^0 S_t^0 + H_t^1 S_t^1.$$

Let $\tilde{V}_t(\phi)$ denote the discounted value process with $\phi \in SF$ (i.e. self-financing). Then, from Ito's differentiation rule

$$\begin{aligned} \tilde{V}_t(\phi) &= V_0(\phi) + \int_0^t H_u^1 e^{-ru} d(S_u^1) \\ &= V_0(\phi) + \int_0^t H_u^1 \sigma e^{-ru} S_u^1 dB_u^\mu. \end{aligned}$$

Lemma 3.2. *The process $\tilde{V}_t(\phi)$ is a martingale.*

3.3. Pricing of European claims. We consider a European claim of the form $f(S_T^1)$.

Definition 3.3. The **investment price** $C(T, f_T)$ for a European claim f_T at time T is the smallest initial investment with which the investor can attain an amount f_T using strategies $\phi \in SF(0)$.

We now show how to price European options.

Theorem 3.4. *Consider an European option with payoff $f(S_T)$ ¹. Then the rational price for the option is*

$$C(T, f_T) = e^{-rT} F(T, S_0^1)$$

¹Satisfying certain integrability condition.

where

$$F(T, S_0^1) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f \left(S_0^1 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} y \right) \right) e^{-y^2/2} dy.$$

Remark 7. Also possible to obtain formula for the **hedge**.

Proof. Observe that

$$C(T, f_T) = \mathbb{E}^\mu(e^{-rT} f(S_T)) = F(T, S_0^1).$$

Using martingale representation theorem one can show that there is a strategy for which the initial investment is $C(T, f_T)$. $\square$

3.4. The PDE approach. Let ϕ be the strategy for which $V_T(\phi) = f_T$ and $V_0(\phi) = C(T, f_T)$. Then, using the Markov property for the process S_t (this is also what we used in the proof of Theorem 3.4)

$$\tilde{V}_t(\phi) = \mathbb{E}^\mu(e^{-rT} f(S_T) | \mathcal{F}_t)$$

Theorem 3.5 (Black-Scholes equation). *Consider a European claim whose payoff is of the form $f(S_T)$ where $f \in C^2$. Its value process $V(t, x)$ satisfies the **linear parabolic PDE***

$$\partial_t V + r S_t^1 \partial_x V + \frac{1}{2} \sigma^2 (S_t^1)^2 \partial_{x^2}^2 V - rV = 0$$

with terminal condition $V(T, S_T) = f(S_T)$.

Some remarks are in order:

- **Existence and uniqueness** of solutions to this PDE with the given terminal condition is standard.
- For $f \notin C^2$ we can approximate by C^2 , **convergence**?

Proof. Approach i) It follows from Ito's differentiation rule that

$$e^{-rt} V_t(S_t^1) - V_0(S_0^1) = \int_0^t e^{-ru} L(V) du + \int_0^t \sigma S_u^1 \partial_x V dB_u^\mu$$

where

$$L(V) = \partial_t V + r S \partial_x V + \frac{1}{2} \sigma^2 S^2 \partial_{x^2}^2 V - rV$$

However, by construction of the value process and since S_t is a martingale, $e^{-rt} V(t, S_t)$ is a martingale. It then follows that $L(V) = 0$ (see Lemma 6.4.2, I think the proof they give is strange but easy to fix).

Approach ii) A portfolio is self-financing iff

$$d(e^{-rt} V_t) = \theta_t d(e^{-rt} S_t) = \theta_t e^{-rt} dS_t - r \theta_t S_t e^{-rt} dt$$

On the other hand

$$\begin{aligned} d(e^{-rt} V_t) &= dV_t e^{-rt} - r e^{-rt} V_t dt \\ &= e^{-rt} (\partial_t V + \mu S_t \partial_S V + \frac{1}{2} \sigma^2 S_t^2 \partial_{S^2}^2 V - r V_t) dt + e^{-rt} \sigma S_t \partial_S V dB_t. \end{aligned}$$

Comparing both expressions $\theta_t = \partial_S V_t$ and BS PDE. $\square$